Modeling and Simulation of Internal Migration using Machine Learning Regression and the Gravity Model of Migration

Abstract

This paper analyzes internal migration patterns and explores how they can be modeled using explanatory features. The goal is to validly depict district level migration flows using causal migration models based on district-level features, for the purpose of analyzing the impact of regional interventions. Thus, sole black-box approaches like time-series extrapolation are not regarded. The models are trained and evaluated for Austria using different time periods as training data and different time lags in features. The models are ranked by how accurately they match the data between 2016 and 2023. Finally, the three best, that is, the extended gravity model of migration, an Autogluon Ensemble model, and constrained gradient boosted trees, are applied to simulate hypothetical migration scenarios involving two selected rural districts in Austria. Compared to other work on the topic of internal migration modeling, we assess not only the model's quantitative accuracy by computing established error metrics on model predictions against the ground truth, but also qualitative behavior by examining the plausibility of simulated intervention scenarios. The most suitable models will be integrated into the agent-based population simulation framework GEPOC in order to investigate the long-term effects of environmental and policy changes on the migration behavior.

CCS Concepts

• **Computing methodologies → Model development and analysis**; *Ensemble methods*; Feature selection.

Keywords

Migration, Simulation, Regression, Machine Learning, Gradient boosting, Gravity Model, Ensemble Trees

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1 Introduction

Migration denotes the change of primary residence of a person. In general, there exist two main types of migration: international migration and internal migration. In international migration, the person changes their location of residence to another country while in internal migration, the movement is within the borders of the same country. On the one hand, internal migration is important for decision-makers, as they have to react to rising or falling population numbers. For example, by planning of new housing areas, jobs, infrastructure, or mobility. On the other hand, internal migration itself can be proactively influenced through interventions or sustainable planning. Examples include actively influencing prices or creating incentives. In general, both sides, internal emigration and immigration can be influenced by targeted policy making. This also differentiates internal from international migration which is not only more difficult, but also more complicated and controversial to influence.

The local impact of migration for a certain region must not be underestimated. High immigration numbers increase the demand for housing, transport, energy, healthcare and can also increase environmental stress. In contrast, high emigration can lead to economic decline of regions and can create a snowball effect so that other important infrastructure like schools or medical facilities close due to low demand.

1.1 Related Work

The most well-known and most used model for internal migration is the so-called gravity model of migration introduced by George Kingsley Zipf in 1946 [20]. In this model, locations attract people based on the number of inhabitants in the origin destination and target destination with a penalty based on the distance between the origin and destination. The name of this model originates from the similarity to the rules of gravitational attraction between two masses. The gravity model of migration, despite its simplicity, is still one of the most used and cited models for internal migration [2] [5] [6]. Other models in the literature include the radiation model [18] and the spatial interaction models [9] as well as modern Machine Learning approaches.

The work of [15] investigates internal migration flows between U.S. counties, comparing traditional migration models with machine-learning-based approaches. While their models achieve strong performance, with Common Part of Commuters (CPC) scores of approximately 0.7, the analysis relies primarily on static features, with population being the only dynamic input. Consequently, temporal changes in socioeconomic and environmental factors are not explicitly modeled. Another approach of machine learning to human mobility is the work of Simini et al. [17]. In the work they created Deep Gravity, a deep neural network approach to simulate mobility flow generation.

It should be emphasized that the models mentioned for internal migration are all very abstract or general and are far from being applicable out-of-the-box. The successful use of the models is primarily linked to the careful selection of relevant features/parameters and the acquisition of corresponding high-quality parametrization data.

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1.2 Aim

The current study is a first step towards a valid causal population model for the long-term simulation of internal migration flows in light of environmental changes, especially with regard to climate change, and corresponding intervention scenarios. To this end, the existing agent-based population model GEPOC ABM is to be expanded to include modules for causal internal migration. GEPOC ABM (see, [3] and (preprint) [4]) is an agent-based framework for analysis of population-level research questions. The model is well established and was already used in various fields of applications, ranging from logistics to epidemiology. To this end, its internal migration module solely relies on statistically extrapolated probabilities incapable of depicting effects of local environmental changes or interventions. To capture corresponding dynamics, the model must be extended.

As an initial assessment, the present study aims to examine different internal migration models to find out which of them predict migration flows accurately effectively and also capture plausible and expected push and pull factors of migration. The models are evaluated through intervention-based simulations to assess the plausibility of the learned causal relationships between features and migration, and clarify which of the models is best suited as part of the GEPOC framework.

We will use the internal migration in Austria from 2018 to 2023 as case study. Considering, that all models for internal migration rely on identification of migration-relevant features, identification of these features and corresponding data sources is another goal of the work. We will compare different selections of features for how well a fitted model can be used to predict internal migration patterns in Austria in recent history. In this assessment process, the simulation aims to predict internal migrants for each year based on a given age- and region-dependent population at the beginning of the corresponding year. However, it should be emphasized that the final application of the model as part of GEPOC ABM will be individual-based and dynamic in terms of a discrete-event simulation.

2 Data and Data Analysis

2.1 Ground Truth Data

Ground truth data on internal migration is published openly on annual basis by Statistics Austria [19] and is available since 2002. The Federal Institution Statistics Austria, provides services in the public interest, compiles statistics and also offers high-quality data. For this ground truth data, a change of main residency within Austria is counted as an internal migration movement if the new address is registered for at least 90 consecutive days.

Statistics Austria offers three different datasets with data since 2002: Data on internal emigrants by origin district, sex and single age classes, analogous data on internal immigrants by destination district, and migration-flow data between municipalities by sex, however without age-resolution.

In order to generate a common reference data set that is broken down by origin, destination (both on district level), sex and age, while also satisfying the given marginal distributions, we used a three-dimensional variant of Iterative Proportional Fitting. This method exists since the 1940s where it was already applied to migration data [8] and can be used to iteratively generate matrices with given row and column sums. The entire process, how the reference data was created is described in Report [4].

2.2 Explanatory Feature Data

Features chosen for this work to predict internal migration include population, economic, educational, environmental factors, available living space, and distances. Distances between districts are based on the car distances between centroids of the district geometry and have been queried using the function ors_directions from the package R openrouteservice, which internally queries the OpenStreetMap API. The source of all features besides rental prices, distances and environmental data are official government institutes: Statistics Austria, österreich.gv.at and the Austrian public employment service. Rental prices have been taken from the widely used market place website willhaben.at. The following list serves as an overview of explanatory features used for training and model fitting. Some features in this list are split up into multiple features before training.

Size Features:

- population
- population within age groups
- population density

Economic Features:

- gdp
- average gross income
- number of companies within certain size classes
- number of overnight stays (tourism)

Living Space:

- average rental prices per m^2
- average land price per m^2
- average real estate prices
- number of building permits within the last 5 years
- Share of rental vs. owner-occupied housing

Educational Features:

- Number of Schools
- Number of Universities within 5km of the district centroid

Other Features:

- travel time to next big city by car and train (centroid based)
- % of green spaces of total district area
- Flooding risk
- Average internet access quality

Due to data availability constraints, several features could not be consistently retrieved for years prior to 2018 or after 2023. Consequently, the final dataset used in all experiments is restricted to the period from 2018 to 2023.

2.3 Exploratory Data Analysis

Figure 1 shows the average relative net migration for all districts in Austria from the year 2016 to 2023. Hereby, the average relative net migration $\overline{r_i}$ for district i across all years T is calculated as:

$$\overline{r_i} = \frac{1}{T} \sum_{t=1}^{T} \frac{I_{i,t} - O_{i,t}}{P_{i,t}} \times 100 \tag{1}$$

where $I_{i,t}$ and $O_{i,t}$ denotes the in-migration and out-migration of district i in year t and $P_{i,t}$ the population of district i in year t.

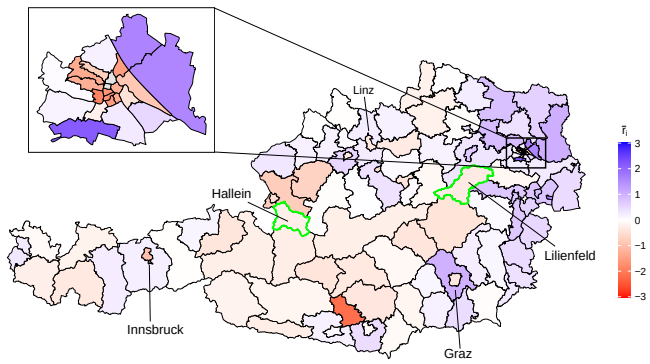

Figure 1: Relative net migration in Austria on a district level, averaged over the years 2016 - 2023

In general, rural areas of Austria show a negative net migration indicating a decline in their population. Interestingly, the big cities Innsbruck, Graz and Linz also experience a negative relative migration whereas their respective surrounding rural districts experience a positive net migration. The zoomed-in area of the figure shows the districts of Vienna where a similar trend can be observed, comparing the inner with the outer districts of the city. In general, one finds, that the rural districts surrounding the major cities and outer districts of Vienna have been the most attractive areas in Austria between 2016 and 2023.

To further analyze migration drivers, Figure 2 presents correlations between district-level features and deviations from a baseline gravity prediction for the year 2022 and age group [20,25). Specifically, we estimate a standard population-based gravity model of migration and derive district-specific push and pull factors by comparing observed in-migration flows to those predicted by the gravity model. The push–pull factor is defined as the logarithmic ratio between observed in-migration and gravity-predicted in-migration 2, capturing whether a district attracts or repels migrants beyond what would be expected based solely on population size and distance effects. Positive values indicate districts that attract more people than predicted (pull factors), while negative values indicate districts that attract fewer people than expected or exhibit net out-migration pressure (push factors). We then compute correlations between these push–pull factors and district features to identify associations between local characteristics and systematic deviations from gravity-based migration expectations.

Figure 2 indicates that, for this age group, some of the selected district-level features exhibit only weak correlations with either high or low pull factors. This lack of pronounced associations suggests that migration behavior in this group is only weakly explained by the available covariates. In practice, this characteristic of the data poses a challenge for model fitting, as predictive models struggle to learn stable and interpretable relationships when systematic feature–outcome associations are weak.

$$\phi_i = \log\left(\frac{M_{in,i}}{\hat{M}^{grav}_{in,i}}\right), \tag{2}$$

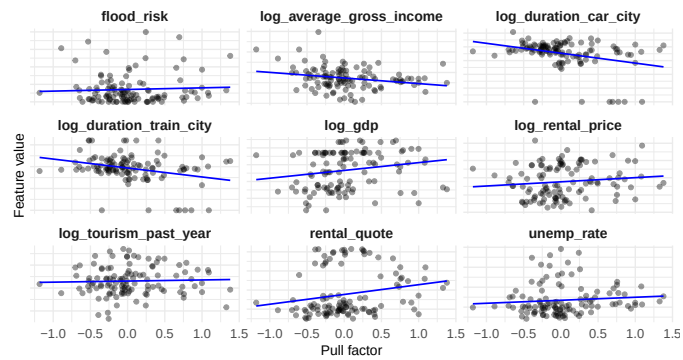

Figure 2: Correlations of district features with the gravity residual pull factor, evaluated for the year 2022 and the age group [20,25)

3 Models and Intervention Scenarios

We furthermore present the different investigated models for internal migration analysis.

3.1 Gravity model

Zipf's gravity model [20], which was first published in his work "The $\frac{P_1 P_2}{D}$ Hypothesis on the Intercity Movement of Persons", has been one of the most used models for migration prediction. The name of the model originates from the similarity to Newton's Law of Gravitation, in the sense that migration scales inversely with distance to some power. To get the number of migrants between two locations i and j, the gravity model usually is written down the following way:

$$M_{ij} = \frac{P_i^{\beta_1} \cdot P_j^{\beta_2}}{D_{ij}^{\beta_3}} \tag{3}$$

When applying the logarithm the formula becomes a linear model that is easy to fit.

The model can be extended with other explanatory variables so that the model is not only depended on the population. This can be done in the following way [2]:

$$\log(M_{i,j,t}) = \beta_1 D_{ij} + \beta_2 P_{i,t} + \beta_3 P_{j,t} + \sum_{k=1}^{p}\left(\gamma_k X^k_{i,t} + \delta_k X^k_{j,t}\right) \qquad \forall\, i \neq j \tag{4}$$

Where P_i and P_j denote log transformed Population, D_{ij} the distance between the districts i and j, X_i and X_j denote the feature vectors of the two locations of interest i and j and $X_{i,t}, X_{j,t} \in \mathbb{R}^{n\times p}$ with n observations and p parameters for a given year t. The terms $\beta_1, \beta_2, \beta_3$ as well as γ_k and δ_k denote the fitted coefficients. This model is fitted using the *glm* (Generalized Linear Models) method

from the statsmodels package [14] and non-informative predictors were removed via AIC-based stepwise selection.

3.2 Constrained Gradient Boosted Decision Trees

Gradient boosting techniques were initially introduced by Jerome H. Friedman [12]. The idea of these models is to iteratively add predictors that iteratively correct the errors of the previous models by minimizing the residual errors. XGBoost and CatBoost [7] [10] are open source implementations of the tree boosting algorithms. These libraries also enables the specification of monotonic constraints for training. The constraint can be set to three values: 1, 0, -1. For 1 and -1 the model will enforces global increase or decrease on the response variable such that

$$f(x_1, x_2, \ldots, x, \ldots, x_{n-1}, x_n) \leq f(x_1, x_2, \ldots, x', \ldots, x_{n-1}, x_n) \quad (5)$$

whenever $x \leq x'$ is an *increasing* constraint; or

$$f(x_1, x_2, \ldots, x, \ldots, x_{n-1}, x_n) \geq f(x_1, x_2, \ldots, x', \ldots, x_{n-1}, x_n) \quad (6)$$

whenever $x \leq x'$ is a *decreasing* constraint.

For example, for training, we can force the model to increase the in-migration count if the variable *income* increases. For the value 0 (default), no constraints are specified and the model can learn freely from the given data.

This monotonic constraint feature of XGBoost and CatBoost is interesting for the task of predicting internal migration to enforce strong prior beliefs of pull and push factors for different age groups in the model.

The constraints used for this model will be published alongside the code and data for the full version of the paper.

3.3 AutoGluon

AutoGluon is an open source AutoMl [11] learning framework that was found to outperform a majority of other Machine learning methods consistently [16]. The description of the training process on their website states: "training a variety of different models, using bagging when training those models, and stack-ensembling those models to combine their predictive power into a “super” model". When applied the data of this work the resulting Autogluon model creates an ensemble of multiple classifiers of the model classes: CatBoost, ExtraTreesMSE, Kneighbors, LightGBM, NeuralNet, RandomForestMSE, WeightedEnsemble, XGBoost. For this work the main hyperparameter that was tuned was the max training time which was set to four hours per fit year span.

3.4 Discrete Choice Model

Additionally to the models that are presented in detail we also ran our experiments with a discrete choice model inspired by the work of [13]. In this model, for a person living in district i within a given age group, the utilities $U_{i,j}$ towards all other destination districts j are computed based on both origin-specific features of district i and destination-specific features of district j as well as dyadic features like distance. Migration flows are then derived from the probabilities $P_{i,j}$ based on these utilities using the formula 7. However, this model has not yet achieved results comparable to the more advanced models discussed later in the paper.

$$P_{ij} = \frac{e^{(U_{ij})}}{\sum_{k=1}^{J} e^{(U_{ik})}} \quad (7)$$

3.5 Model Training and Fitting Setup

The available full data spans the years 2018 to 2023. In terms of training and fitting some interesting questions arise. Since migration is a complex phenomenon it is not yet answered in the literature what an appropriate time lag for all features are. Additionally, push and pull factors in migration can change over time and are depended on factors like economy or major events like the COVID-19 pandemic.

To get around these problems, the models have been trained on different subsets of data and time lags. To avoid overfitting, the models are fitted and trained on four and three year windows of data and are evaluated on two years of unseen validation data. Model performance on this validation set is used to determine whether training on three or four years of historical data yields better generalization.

As a preprocessing step, all count-based variables were normalized by the population size of the respective spatial unit. For example, target-district school counts were converted into schools per 1,000 inhabitants of the target district's population. This population-normalization avoids scale effects and allows the model to capture relative availability rather than absolute counts.

Training and fitting procedures differed across model classes. The gravity model was estimated using Python statsmodels generalized linear models, with feature selection performed via a stepwise Akaike Information Criterion (AIC) [1] procedure that adds variables only if they improve model predictions while penalizing model complexity.

The two Gradient Boosting models were trained using a standard scikit-learn grid search with five-fold cross-validation to identify optimal hyperparameters. The search optimized L2 and L1 regularization strength, column sub sampling rate, tree depth, learning rate, and the number of estimators. As described in the discussion section, the column subsampling was conducted to reduce the feature importance of population columns. The defined evaluation metric was set to negative log-likelihood.

For AutoGluon, model training had a fixed time budget of four hours, with the ensemble configuration comprising two stacking levels and five bagging folds.[11]. The chosen evaluation metric for this training process was the CPC score 11.

For training, the models predict migration flows between all pairs of districts, providing a more granular representation than models that simulate net migration or aggregate inflows and outflows. Predicting flows for all direction pairs among the 116 districts (i.e., $116 * 115$) for 18 age groups allows us to predict and simulate migration across all of Austria.

The evaluation of the predictive accuracy of the different models between fit years was done by comparing the mean relative error $\epsilon_{rel,j}$ of the inbound migration across all districts.

$$\varepsilon_{\text{rel},j} = \frac{\left| \sum_{t=1}^{T} \sum_{i \neq j} \hat{M}_{ij,t} - \sum_{t=1}^{T} \sum_{i \neq j} M_{ij,t} \right|}{\sum_{t=1}^{T} \sum_{i \neq j} M_{ij,t}}. \tag{8}$$

Here t denotes a given year within the set of all years T for which the model should be evaluated and $\hat{M}_{ij,t}$ denotes the predicted migration flow.

Number of correctly predicted trends Another important measure to evaluate fitted models is the number of correctly predicted net migration signs.

$$S_{agree} = \sum_{i=1}^{n} \mathbb{I}\left[\text{sign}\left(\widehat{\text{I}}_i - \widehat{\text{O}}_i\right) = \text{sign}(\text{I}_i - \text{O}_i)\right], \tag{9}$$

where I_i and O_i denote the sum of inward and outward migration, $\widehat{\text{I}}_i$ and $\widehat{\text{O}}_i$ the predicted sum of inward and outward migration summed for the years 2018 to 2023. The function $\mathbb{I}$ denotes the indicator function which has the value 1 if the sign is identical and 0 if the sign is different.

Common Part of Commuters An established measure for the quality of the predicted district pair flows (OD flows) is the Common Part of Commuters. The common part of Commuters is a special form of the Sørensen–Dice index 10.

$$\text{Dice}(A, B) = \frac{2\,|A \cap B|}{|A| + |B|} \tag{10}$$

Equation 11 shows how to calculate the Common part of Commuters for the given observed flows M_{ij} and predicted flows $\hat{M}_{ij}$.

$$\text{CPC} = \frac{2 \sum_{i,j} \min\left(M_{ij}, \hat{M}_{ij}\right)}{\sum_{ij} M_{ij} + \sum_{i,j} \hat{M}_{ij}} \tag{11}$$

The CPC is a value in the interval [0,1], where 0 does not show any overlap and 1 is a perfect match.

3.6 Simulation of Intervention Scenarios

To analyze the qualitative validity of the models, we investigate the impact of fictional interventions. To simulate migration flows under hypothetical scenarios with changed district features, the metadata of the years 2018 to 2023 is changed and the models are used again to predict migration flows.

We specifically ask, how an intervention could counteract the rural depopulation effect. To test this, the Austrian rural districts Lilienfeld and Hallein have been chosen as region for the intervention, due to their particularly high negative net migration. Both districts are highlighted in green in the map 1.

The models offer the following strategies for defining interventions by varying feature values:

- To simulate economic updraft of a district one can manipulate the number of big companies $c_{<size-class>}$, the GDP gdp and the average income $income$.
- To simulate shorter travel times to the next big city, the feature $t_{urban,car}$ can be manipulated. This value could change if new streets or tunnels are constructed that decrease the time to travel to the next major city.
- To simulate higher education, the number of universities in the district are be changed using the variable uni. Note that currently Hallein and Lilienfeld do not have an university.
- To simulate an increase in the construction of living space the variables $dp_{<size>}$ is manipulated. Also the variable dp_{3-10} denotes the number of permitted dwellings with three to ten rooms within the last five years.

In total, six intervention scenarios S1 to S6 are carried out, each of which intended to increase the net migration in the two districts. Table 1 shows the deltas applied to the described feature variables for the investigated scenarios. Values with + and - are added or subtracted from the original values for each year, whereas * is a factor. Values without an operator are set to the specified value, meaning that for S5 the two districts have no companies in the size class 500 - 1000 employees.

It should be noted that potential cross-effects between the intervention variables were either estimated manually (S1, S5, and S6) or neglected in this study. This applies in particular to effects that occur with a delay as a result of a feedback loop, e.g. higher migration leads to a larger population and thus to higher demand for education.

4 Results

4.1 Training Results

With the exception of the XGBoost models, all approaches achieved their best validation performance (In terms of relative error 8) when trained on four years of historical data. In contrast, the XGBoost models demonstrated best generalization when trained on three years of data. All figures and results related to XGBoost and CatBoost relate to the constrained versions. Using feature level constraints described in section 3.2. Table 2 shows the best model of each model family evaluated based on the criteria described in Equation 8. The Time Lag parameter denotes the temporal offset applied between input features and the target variable during both training and inference. For example, a time lag of one year implies that the model uses data from year t (e.g., 2020) to predict migration flows for year t + 1 (e.g., 2021).

The coefficient of determination (R^2) was used as a measure of the explained variance in the model. The tables 3 4 present the mean R^2 scores ($\overline{R^2}$) for the aggregated Immigration and aggregated Emigration across all age groups as well as the worst and best performing age groups in terms of R^2. The R^2 scores in these tables are measured for the predictions of unseen data during training (=validation set). Specifically, for a model trained on the period 2019–2022, R^2 scores are evaluated using predictions for the years 2018 and 2023. This setup assesses the model's ability to generalize both backward and forward in time beyond the training window.

Figure 3 shows the CPC scores of the trained models evaluated on the validation dataset.

Although the AutoGluon model achieves a lower mean relative error and a better CPC score, we consider the constrained XGBoost model as the best resulting model of this study. The primary reason for this choice is the interpretability introduced through the imposed monotonicity constraints, which provide a semi–white-box

Table 1: Changes of feature values for the definition of various scenarios

Scenario	$c_{500-999}$	c_{1000+}	gdp	$income$	uni	$t_{urban,car}$	dp_{3-10}	dp_{11+}
S1 - stronger economy	+3	+2	*1.1	*1.2				
S2 - shorter distance						-20		
S3 - university					+1			
S4 - S1 + S2 + S3	+3	+2	*1.1	*1.2	+1	-20		
S5 - weaker economy	0		*0.85	*0.8				
S6 - increased living space							+10	+15

Table 2: Model performance summary

Model	Fit Years	Time Lag	$\overline{\varepsilon_{\text{rel},j}}$	S_{agree}
Gravity	2019 - 2022	1 year	0.25	86/116
XGBoost	2020 - 2022	1 year	0.14	90/116
CatBoost	2020 - 2023	0 years	0.25	98/116
Autogluon	2019 - 2022	0 years	0.09	97/116

Table 3: Comparison of R^2 statistics for In-Migration across models and age groups

Model	$\overline{R^2}$	Min R^2	Age (Min)	Max R^2	Age (Max)
Gravity	0.80	0.54	[85,90)	0.91	[25,30)
Autogluon	0.95	0.78	[80,85)	0.98	[25,30)
XGBoost	0.91	0.70	[85,90)	0.97	[25,30)
CatBoost	0.91	0.75	[85,90)	0.98	[20,25)

Table 4: Comparison of R^2 statistics for Out-Migration across models and age groups

Model	$\overline{R^2}$	Min R^2	Age (Min)	Max R^2	Age (Max)
Gravity	0.82	0.62	[85,90)	0.91	[35,40)
Autogluon	0.95	0.86	[85,90)	0.98	[25,30)
XGBoost	0.93	0.76	[85,90)	0.97	[25,30)
CatBoost	0.93	0.75	[80,85)	0.98	[45,50)

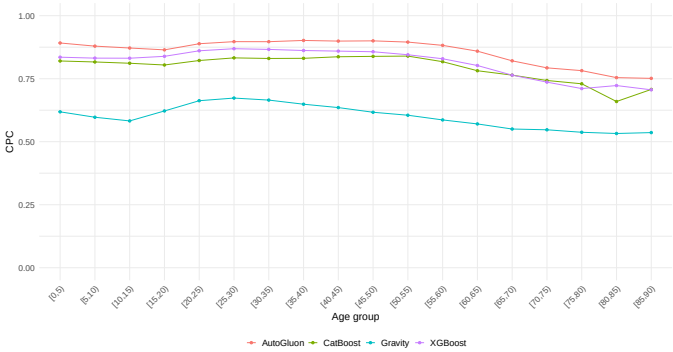

Figure 3: CPC Scores of different models on validation data for different age groups

structure. Furthermore, the constrained model produces more plausible behavioral responses in the intervention scenarios discussed in Section 3.6.

Figure 4 presents the SHAP values of the best-performing constrained XGBoost model for the age group [20,25), illustrating the relative importance and directional effects of the features.

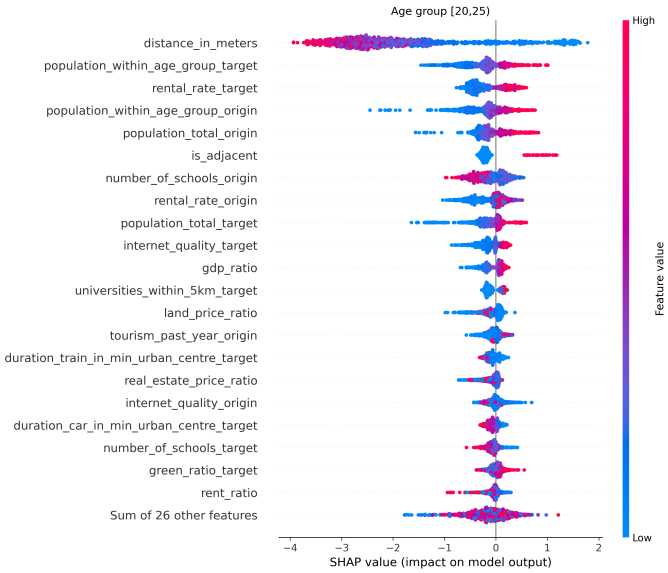

Figure 4: SHAP values for the best constrained XGBoost model for the age group [20,25)

SHAP values indicate how each feature contributes to the predicted migration flow from origin to target. Positive values increase the predicted number of migrants, while negative values decrease it. For example, greater distance between districts (distance_in_meters) contributes negatively to migration flows. Features are ordered on the y-axis by their overall importance, measured as the mean absolute SHAP value.

4.2 Intervention Scenario Results

Figures 5 and 6 show the results of all simulation results compared to the prediction of the original data as well as the observed value. The y-axis displays the predicted value for the net migration in the given year (inward migration - outward migration). The expected results for S1 to S4 is a higher net migration, since a better economy

and a university are considered a pull factor. S5 should lower net migration, since an economic downtrend is expected to be a push factor. Scenario S6 is more interesting, since the construction of new buildings alone might not be enough to increase inward migration without additional factors to make the region more attractive. For the AutoGluon model and Gravity Model of Migration the intervention scenarios often show unreasonable results. In both models a weaker economy yields more net migration while a shorter distance to urban hubs makes the districts also less attractive.

For the constrained GBDT model, all scenario results conform to the expected prior assumptions for both districts. This behavior is enforced by applying monotonic constraints prior to training. Compared to the other models, this approach produces the most pronounced scenario effects. In future work, domain experts will be consulted to assess whether the magnitude and direction of these predictions are realistic. Although these effects cannot be proven in the absence of ground-truth counterfactuals, we posit that incorporating monotonic constraints yields more plausible and potentially more reliable scenario responses than models that learn unrestricted relationships solely from historical migration data.

5 Discussion and Limitations

Although the models achieve promising results in terms of prediction errors, as well as R^2 scores, not all of the learned causal relationships look plausible. Moreover, depending on the model, the correct migration trend is captured in 86 to 98 of the 116 districts in the validation sets, indicating a substantial spatial misprediction.

One of the main challenges of this study is the disproportionate influence of certain features on model predictions. Population size and distance explain a large share of the variance in the data and therefore dominate the prediction of migration flows from origin to destination. In contrast, variables such as green space availability, economic indicators, or flood risk exhibit consistently low feature importance. As a result, all evaluated model archetypes tend to overfit these dominant variables and fail to learn meaningful relationships for the remaining features.

This issue is further exacerbated by the high degree of economic homogeneity across Austrian districts. As illustrated by the correlation analysis (see Figure 2), most features show only weak associations with district attractiveness. Several mitigation strategies were explored, including modeling relative migration flows to emphasize attractiveness rather than absolute counts and removing population as an input feature. However, both approaches substantially degraded predictive performance.

The most effective approach identified in this study was the use of feature constraints combined with column sampling during training. These techniques reduce reliance on dominant variables and encourage the model to incorporate a broader set of features. Importantly, our trained constrained models do not claim to reveal causal mechanisms or guarantee that the imposed scenario assumptions reflect real-world migration behavior. Rather, they serve to improve the plausibility and interpretability of model outputs in the presence of noisy and incomplete data.

Unobserved factors may still drive migration patterns, such that economically improving districts or areas with increasing green space experience negative net migration. Unconstrained models are unable to account for such latent effects and therefore fail to adjust migration flows accordingly. While feature constraints introduce additional bias, they provide a principled way to align model behavior with domain knowledge and to generate scenario outcomes that are more plausible than those learned purely from noisy data.

One major limitation is also the absence of some important features, due to both data limitations and time constraints. Additional features that could improve the model predictions are the number of vacant dwellings, the development of public transportation, attractive amenities, accessibility to healthcare, availability of childcare or employment distribution across economic sectors. While data for some of these features is available, they have not yet been integrated.

While the trend of the simulation scenarios looks plausible, there has been no deeper investigation of their numerical accuracy. In further research, it is the goal to discuss the predicted delta migration in the simulations with domain experts to see how realistic they are and to compare them to other existing models.

Solving the task of migration simulation using regression also has some inherit disadvantages. When predicting flows in a tabular manner, some more complex relations are not captured. For example: If a region gets more attractive than other neighboring regions, the regression based simulation will pull more people from the neighboring districts but will not reduce migration into neighboring districts which can occur in reality. To address spatial spillover effects, we propose two methodological enhancements. The first approach utilizes Spatially Lagged Explanatory Variables (SLX), where the attractiveness of a destination j is modeled as a function of its own attributes and the weighted attributes of its neighbors. For a given feature X (e.g., GDP), the spatial lag could be calculated as:

$$SLX_j = \sum_{k \neq j} w_{jk} X_k \tag{12}$$

where w_{jk} denotes the elements of a spatial weight matrix defining the connectivity between destination j and neighbor k. This allows the model to capture direct cross-border benefits.

The second approach incorporates a measure of Regional Accessibility (often termed *centrality* in Competing Destinations frameworks) to capture broader agglomeration context. We calculate an accessibility index A_j for every destination:

The second approach that could resolve this issue adopts a Spatially Smoothed Utility perspective, acknowledging that the total utility derived from a location extends beyond administrative boundaries. We could define the effective utility $\tilde{V}_j$ of a destination as a distance-weighted aggregate of the utilities of the surrounding region:

$$\tilde{V}_j = \sum_{k} w_{jk} V_k \tag{13}$$

where V_k represents the intrinsic utility (attractiveness) of location k and w_{jk} is a spatial weight that decays with distance. By substituting the isolated local utility with this spatially smoothed value, the model accounts for the "regional" benefit of a destination, capturing positive spillovers from highly attractive neighbors.

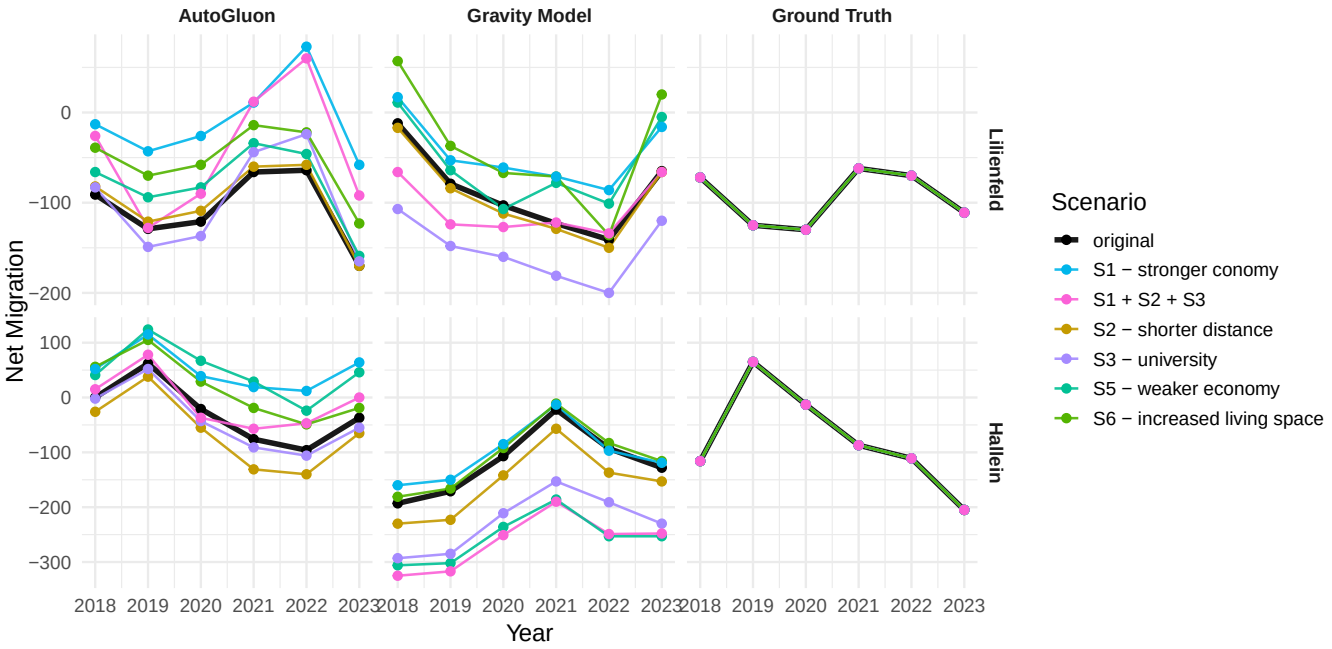

Figure 5: Predicted net migration of the AutoGluon and Gravity Model for various simulation scenarios in two example districts

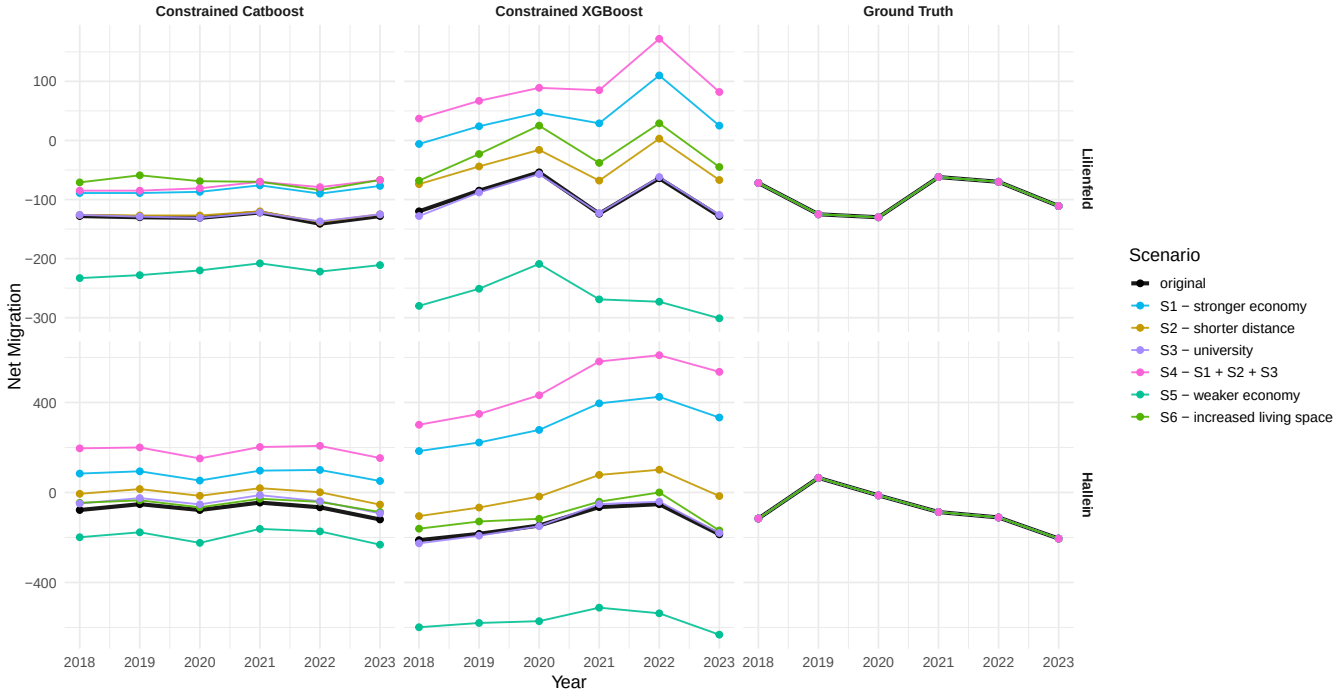

Figure 6: Predicted net migration of the Constrained CatBoost and XGBoost models for various simulation scenarios in two example districts

Finally, it should be noted that the current modeling approaches map the regions' features onto internal migration purely unidirectionally. Thereby they disregard causal relationships between the features, e.g., between the various economic variables, as well as

potential feedback effects of migration figures on the features, e.g., rising immigration usually leads to rising rental prices. While neglecting feedback loops is reasonable for the current study, because the models only simulate one year into the future, however, these effects must be taken into account when integrating the models into the GEPOC framework.

6 Conclusion

The work in this paper shows that Zipf's extended Gravity Model for migration, Tree Ensemble models as well as constrained Gradient Boosting Trees can be used to predict migration flows within Austria with average relative errors of 25%, 14% and 9% for in-migration flows. The quality of the fit is clearly not only due to the models themselves, which have already been applied successfully several times in international literature, but above all to the sensible selection of features and feature data that (a) cover the most relevant motives for migration of Austrians, (b) are suitable for simulating intervention scenarios, and yet (c) avoid overfitting effects.

We also showed that constrained Gradient Boosted Trees are capable of incorporating prior beliefs about causal relationships between internal migration and explanatory variables, while unconstrained models frequently infer implausible patterns — an issue amplified by Austria's homogeneous context and the inherent noise in migration data. Future research on these regression based internal migration models should include comparison of simulation results with past recorded economic shocks to evaluate the plausibility of the simulations.

Internal migration is a highly individual issue, and the reasons why people migrate within a country vary greatly from country to country. This is primarily due to different lifestyles, cultural, geographical, or climatic differences. Features that may be considered highly important for quality of life by the population of one country may be unimportant in another. On the one hand, this means that the models with their current selection of features cannot be universally applied to other countries. In Austria, for example, crime can be ruled out as a motive for migration. The corresponding rates are minimal and there are hardly any significant regional differences. It can be assumed that this feature is certainly relevant to validly depict migration in other countries. On the other hand, for countries where indeed similar features are considered relevant, such as other European countries, the models can be used to better understand differences in migration priorities by comparing the parameters of the trained models. The biggest challenge here is undoubtedly the acquisition and processing of migration and feature data. Finding and processing this data in acceptable quality for Austria alone, the authors' home country, proved to be a surprisingly big challenge.

The next steps involve integrating the evaluated models into the GEPOC ABM population simulation framework. Even though Gradient Boosted Trees proved to be the most accurate model in the presented study, the models will face new challenges when integrated. Since in GEPOC ABM the models will no longer evaluate by simple multiplications with the population figures but need to generate rates and probabilities on the individual level, the assessment must be reevaluated. Factors such as flexibility and computational performance will become even more important. Feedback between migration figures and features must be taken into account as well, e.g. by incorporating System Dynamics features.

7 Source Code and Data

To keep the author anonymous for review the repository is not yet linked in this paper. The source code as well as the data and description of data sources used in this study will be included in the full version of the paper.

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